

# Final Presentation for ISIT'25 Hackathon

## QNNs for multi-level entanglement classification

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## Why multi-level entanglement classification matters?

- **Quantum Communication & Networking**
  - Dynamic routing: classify link quality (low/med/high entanglement) to decide when to swap or purify
  - Resource allocation: conserve scarce entangled pairs by only using “high-grade” links for critical tasks
- **Quantum Key Distribution (QKD)**
  - Security tuning: higher-entanglement states yield stronger correlations and lower quantum bit-error rates
  - Adaptive protocols: switch between BB84, E91 or CV-QKD based on available entanglement level
- **Quantum Sensing Applications**
  - Noise resilience: discard low-entanglement probes that contribute more noise than signal
- **Resource-Efficient Quantum Computing**
  - Error mitigation: pre-classify states to decide when to trigger error-correcting routines
- ***Problem Statement : Classifying entanglement into actionable tiers, we transform a binary ‘entangled/not-entangled’ check into a continuous resource management tool***

## Why QNN is the way forward?

- **Hilbert-Space Embedding**

- Quantum Feature Maps: map density matrices to exponentially large feature spaces
- Non-trivial data patterns: capture complex entanglement indicators that classical kernels struggle with

- **Trainability Aspects**

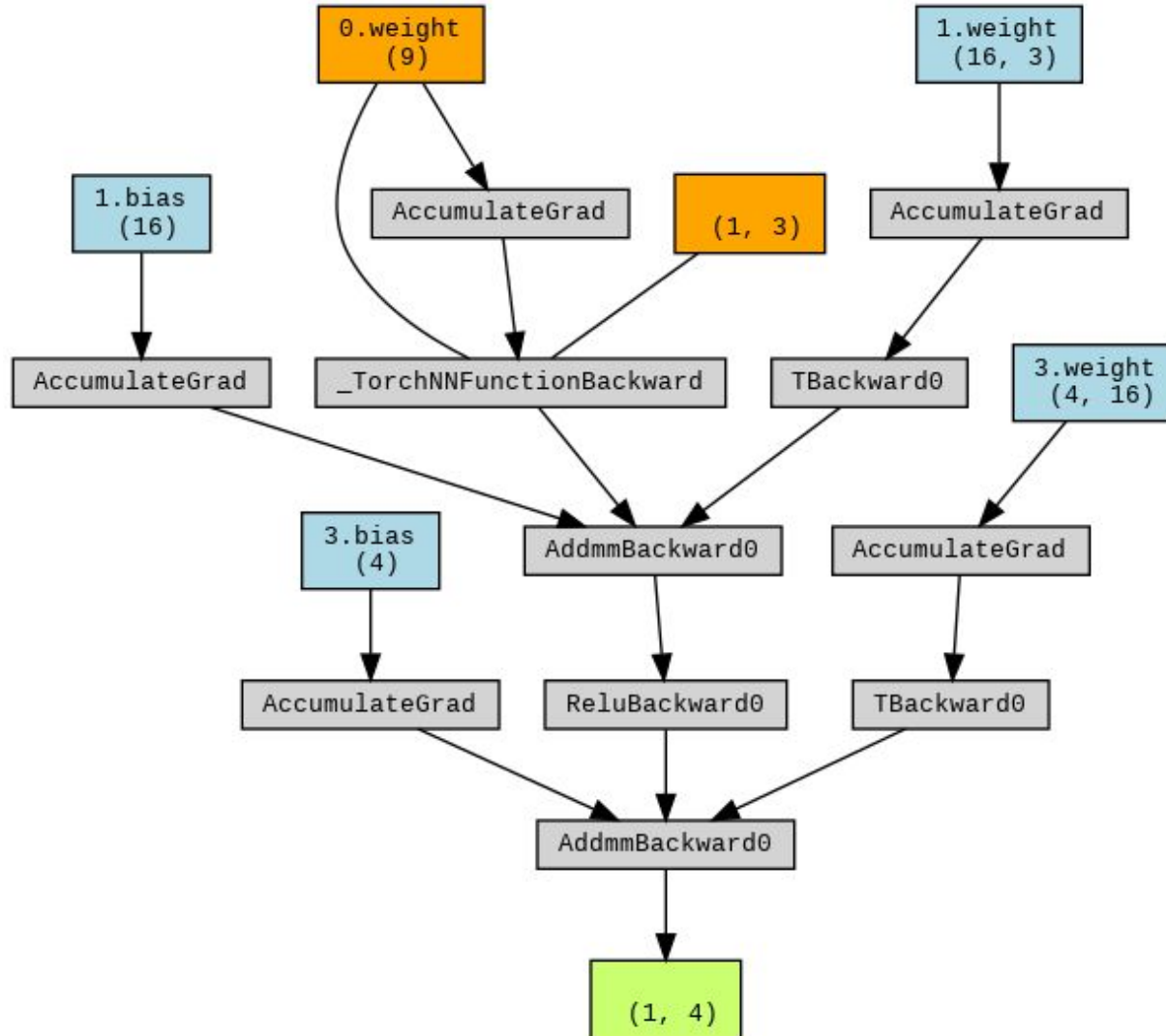
- **Hybrid Training:** classical optimizer steers quantum circuit parameters, enabling few-shot learning on small datasets
- **Hardware-Efficient Ansatz:** exploit native gates, reducing depth and error accumulation
- **Noise-Aware Training:** incorporate noise models into loss-function, boosting real-device performance

- **Transferability:** prototypes on 2-qubit systems can be extended (with minimal retraining) to higher-dimensional systems

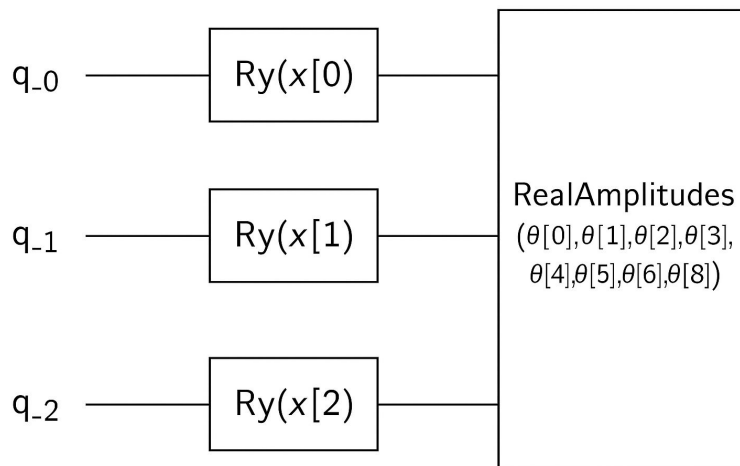
## Dataset Generation and Noise modelling

- Bell Diagonal States (BDS) are common in experimental scenario.
  - BDS are convex combination of the four Bell states.
  - PPT criterion: BDS are entangled if all mixing weights  $< 0.5$ ; entangled if one of the weights  $> 0.5$
- Entanglement was quantified using Negativity.
- We generated a labelled dataset:
  - $\langle \text{density matrix}, \text{label}, \text{exp\_XX}, \text{exp\_YY}, \text{exp\_ZZ} \rangle$
  - exp\_XX is the ideal expectation value of the pauli X observable
  - **Label grades** - 0: separable, 1: low, 2: medium, 3: high
- Checks and Balances
  - Dataset is balanced and equally populated with all four label grades.
- Training: Model is trained on the ideal expectation values
- Testing
  - Four noisy channels are considered - Phase damping , Amplitude Damping , Correlated amplitude damping , Correlated phase damping.
  - Noise is added to the pauli expectation values and the model predicts whether or not there is a change in gradation of entanglement.

# Hybrid Quantum Neural Network - Architecture



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```

observables = [
    Pauli("ZII"), Pauli("IZI"), Pauli("IIZ"), # Individual
]

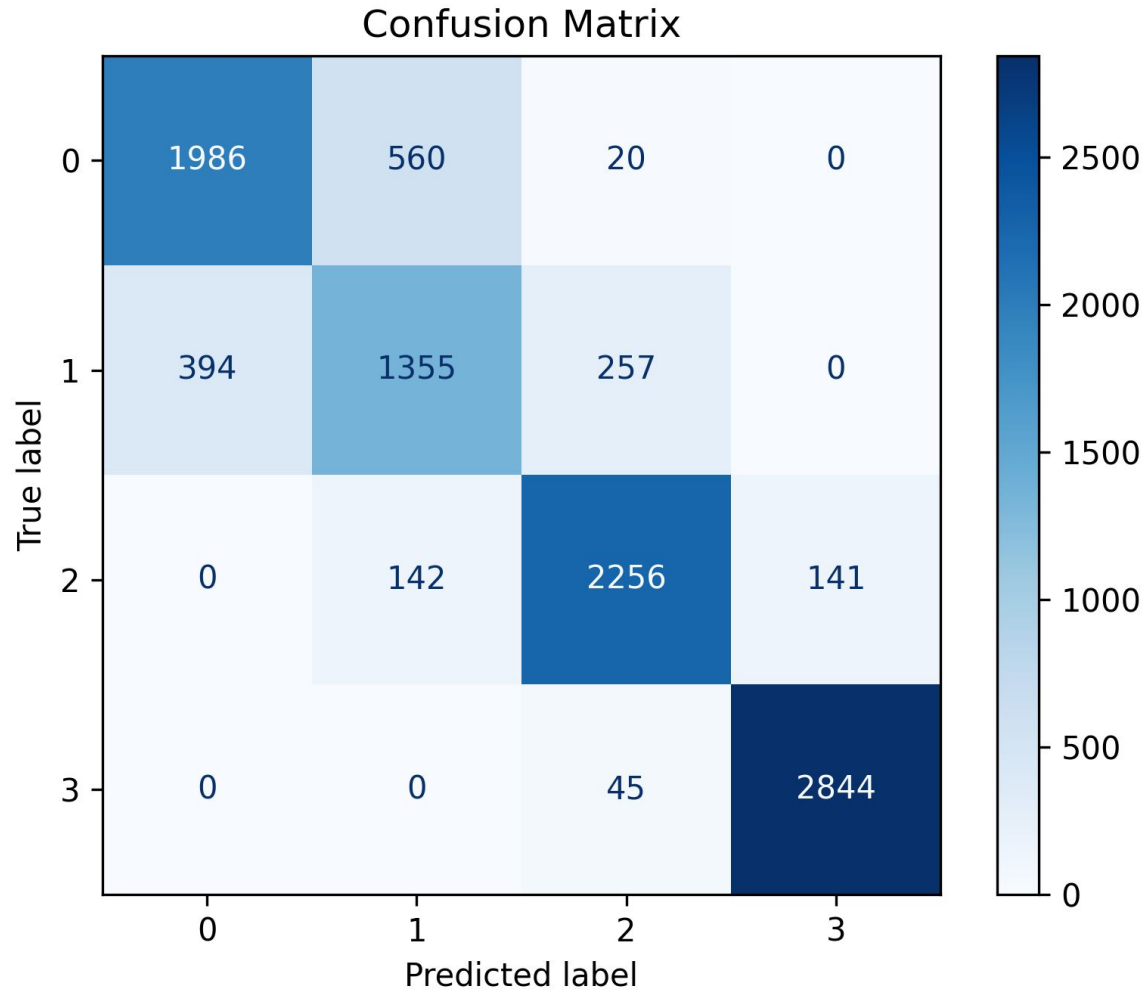
def create_qnn(num_qubits: int = 3):
    x = ParameterVector("x", num_qubits)
    circuit = QuantumCircuit(num_qubits)
    for i in range(num_qubits):
        circuit.ry(x[i], i)
    ansatz = RealAmplitudes(num_qubits=num_qubits, reps=2, entanglement="linear")
    circuit.compose(ansatz, inplace=True)

    return EstimatorQNN(
        circuit=circuit,
        input_params=x,
        weight_params=ansatz.parameters,
        estimator=Estimator(),
        input_gradients=True,
        observables=observables
    )

qnn = create_qnn()
model = TorchConnector(qnn)

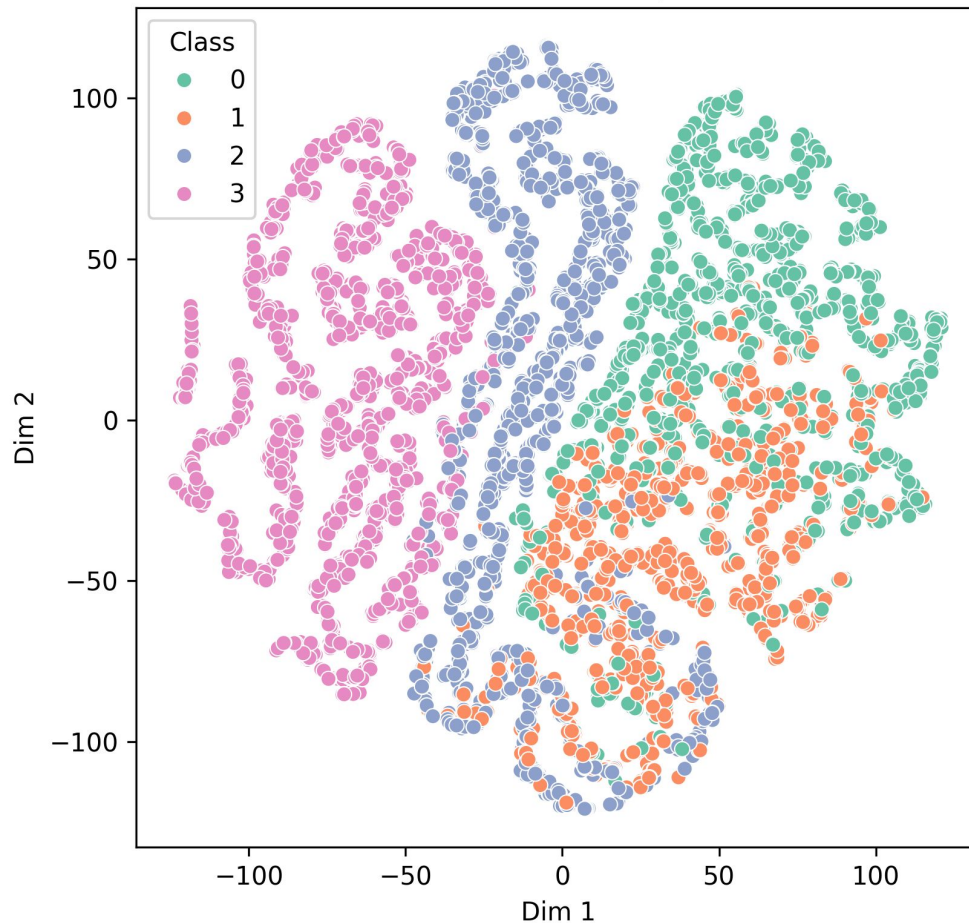
# Full classifier
classifier = nn.Sequential(
    model,
    nn.Linear(3, 16, bias=True),
    nn.ReLU(),
    nn.Linear(16, 4, bias=True),
)
  
```

# Hybrid Quantum Neural Network - Results

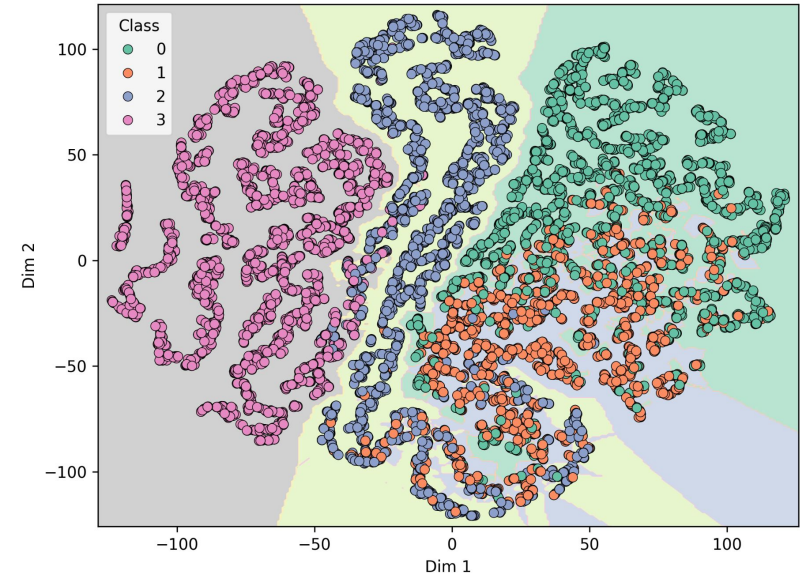


# Hybrid Quantum Neural Network - Results

t-SNE of QNN Outputs

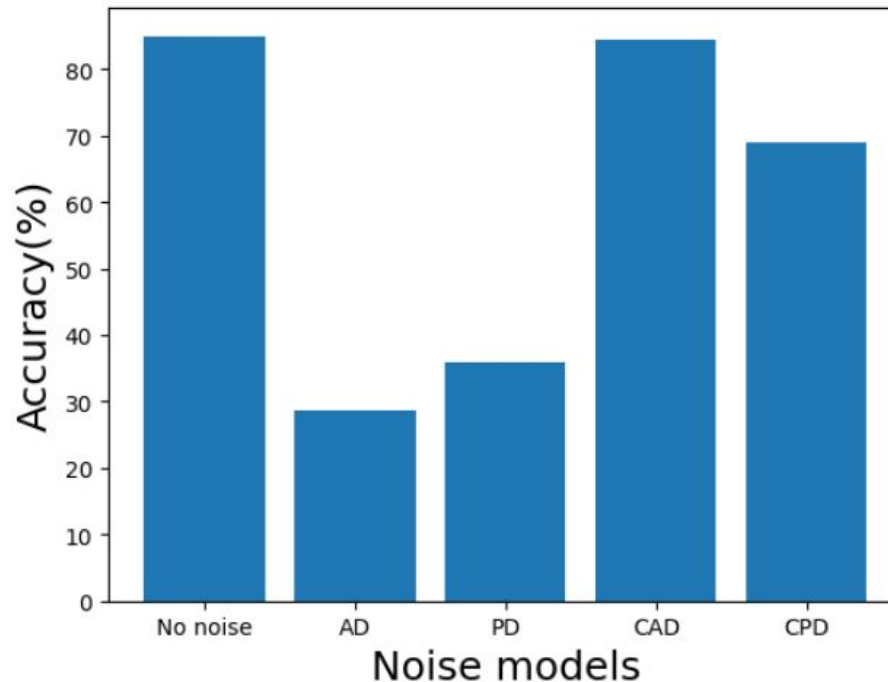


t-SNE with Decision Boundaries





# Alice Bob Noisy Communication Experiment



The accuracy dropped severely than ideal case for AD and PD noise models indicating that model can predict existence of those noise in a quantum internet.

For detecting correlated noises (CAD and PAD) which potentially increase entanglement we might require a finer classification for levels of entanglement or a better model.

# Alice Bob Noisy Communication Experiment

